

The derived types are also known as structured types. The derived types include pointers to other types, arrays of other types, function types, structures, and unions.

Fundamental Types

The fundamental types represent types that cannot be divided into more basic elements, and are the model for representing elementary data on machine level. The fundamental types are sometimes referred to as unstructured types, and are used as elements in creating more complex derived or user-defined types.

Arithmetic Types

The arithmetic type specifiers are built from the following keywords: void, char, int, float, and double, together with prefixes short, long, signed, and unsigned. From these keywords you can build the integral and floating-point types.

Integral Types

Types char and int, together with their variants, are considered integral data types. Variants are created by using one of the prefix modifiers short, long, signed, and unsigned.

The table below is the overview of the integral types – keywords in parentheses can be (and often are) omitted.

The modifiers signed and unsigned can be applied to both char and int. In the absence of unsigned prefix, signed is automatically assumed for integral types. The only exception is the char, which is unsigned by default. The keywords signed and unsigned, when used on their own; mean signed int and unsigned int, respectively.

The modifiers short and long can be applied only to the int. The keywords short and long used on their own mean short int and long int, respectively.

Type	Size in bytes	Range
(unsigned) char	1	0 to 255
signed char	1	- 128 to 127
(signed) short (int)	1	- 128 to 127
unsigned short (int)	1	0 to 255
(signed) int	2	-32768 to 32767